

which is non-negative via the above observation that $\frac{\partial \pi_i}{\partial r} > 0$. Similarly, for all $r' \geq r$:

$$\begin{aligned} [U_i(\mathcal{S}, r') - U_i(\mathcal{D}, r')] - [U_i(\mathcal{S}, r) - U_i(\mathcal{D}, r)] &= [U_i(\mathcal{S}, r') - U_i(\mathcal{S}, r)] + [U_i(\mathcal{D}, r') - U_i(\mathcal{D}, r)] \\ &= (u + c)(\pi_i(r') - \pi_i(r)) + \tilde{u}(\pi_i(r') - \pi_i(r)), \end{aligned}$$

which is non-negative via the same observation. Finally, for all $r' \geq r$:

$$[U_i(\mathcal{I}, r') - U_i(\mathcal{D}, r')] - [U_i(\mathcal{I}, r) - U_i(\mathcal{D}, r)] = U_i(\mathcal{D}, r') - U_i(\mathcal{D}, r) = \tilde{u}(\pi_i(r') - \pi_i(r)),$$

which, again, is non-negative. Thus, via Topkis's monotone comparative statics theorem (see [Topkis \(1998\)](#)), there is uniformly more sharing. ■

Proof of Proposition 2. The social media game is supermodular and has increasing differences in sensationalism and (the negative of) reputational concerns. To see this, note for all $\kappa' \geq \kappa$ and $d' \leq d$:

$$[U_i(\mathcal{S}, \kappa', d') - U_i(\mathcal{I}, \kappa', d')] - [U_i(\mathcal{S}, \kappa, d) - U_i(\mathcal{I}, \kappa, d)] = U_i^{(2)}(\kappa', d') - U_i^{(2)}(\kappa, d) = (\kappa' - \kappa)S_i + (d - d')D_i$$

which is non-negative. Moreover, note that comparing $\mathcal{S}$ and $\mathcal{D}$ is identical to comparing $\mathcal{S}$ and $\mathcal{I}$ because parameters (κ, d) affect both $\mathcal{I}$ and $\mathcal{D}$ identically (they only factor into the payoff of action $\mathcal{S}$). For this same reason, we note that $[U_i(\mathcal{I}, \kappa', d') - U_i(\mathcal{D}, \kappa', d')] - [U_i(\mathcal{I}, \kappa, d) - U_i(\mathcal{D}, \kappa, d)] = 0$. Thus, via Topkis's theorem (see [Topkis \(1998\)](#)), there is uniformly more sharing. ■

A.3 Proofs from Section 4

Proof of Lemma 1. To obtain a contradiction, suppose that there exists an agent i and an agent j with $\ell_i = \ell_j$ but either (i) $b_i^* \neq b_j^*$ or (ii) $b_i^{**} \neq b_j^{**}$.

Without loss of generality, suppose that $b_i^* < b_j^*$. By way of contradiction suppose $b_i^{**} > b_i^*$, and consider priors $\tilde{b} \in (b_i^*, \min\{b_j^*, b_i^{**}\})$ where agent i would ignore but agent j with that same prior would dislike. However, both agents with prior $\tilde{b}$ receive payoff $\tilde{u}(1 - \pi(\tilde{b})) - \tilde{c}$ from disliking and payoff 0 from ignoring. Thus, one of them must not be playing a best response, and $b_i^{**} = b_i^*$.

Thus, when agents i and j both have some prior $b' \in (b_i^*, b_j^*)$, agent i shares and agent j dislikes. By symmetry of agent i and j 's network positions, it is clear that for agent i and agent j with prior b' that $U_j(\mathcal{S}) - U_i(\mathcal{S}) = p_s(\kappa + d)$. Similarly, $U_j(\mathcal{D}) - U_i(\mathcal{D}) = 0$. But in this case,

$$[U_j(\mathcal{S}) - U_i(\mathcal{S})] - [U_j(\mathcal{D}) - U_i(\mathcal{D})] = [U_j(\mathcal{S}) - U_j(\mathcal{D})] + [U_i(\mathcal{D}) - U_i(\mathcal{S})] = p_s(\kappa + d) > 0.$$

This implies that either $[U_j(\mathcal{S}) - U_j(\mathcal{D})] > 0$ or $[U_i(\mathcal{D}) - U_i(\mathcal{S})] > 0$ (or both). This yields a contradiction because at prior b' , it is supposed to be a best response for agent j to play $\mathcal{D}$ and a best response for agent i to play $\mathcal{S}$. Thus, $b_i^* = b_j^*$.

Without loss of generality, suppose that $b_i^{**} < b_j^{**}$. If $b_i^{**} \leq b_j^*$, then for priors $b'' \in (b_i^{**}, b_j^*)$, agent i